GITAM University

Subject-Physics

Topic: Magnetic materials

Sayan Chaudhuri

Magnetic Field → H → A/m in SI

Magnetization → M → A/m in SI

Magnetic Susceptibility → M → A/m in SI

Magnetic Induction → B → T or Wb/m^2 in SI

$$M \propto H$$

$$M = \chi H$$

$$B = \frac{\phi}{A}$$

ϕ is magnetic flux

$$B = \mu_0 (H + M)$$

$$B = \mu_0 (1 + \chi) M$$

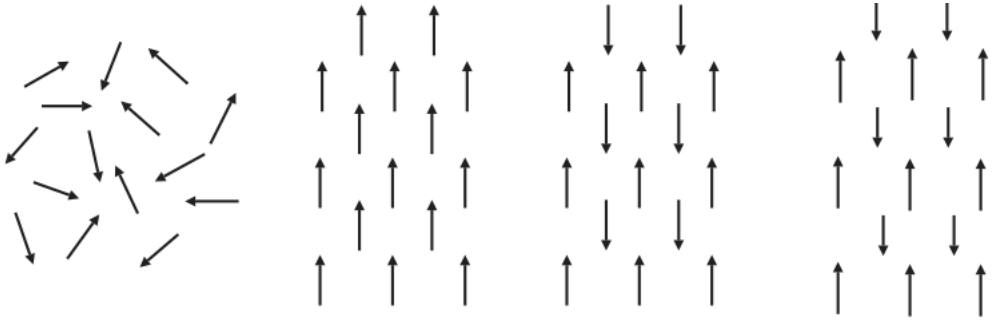
$$B = \mu H \rightarrow \mu = \mu_0 (1 + \chi)$$

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi$$

$\mu \Rightarrow$ Absolute permeability of the material

Unit ➔ Henry per meter (H/M)

$\mu_r \Rightarrow$ relative permeability of the material



Diamagnetic material $\mu_r < 1$

Paramagnetic material $\mu_r > 1$

Ferromagnetic material $\mu_r \gg 1$

ORIGIN OF MAGNETIZATION

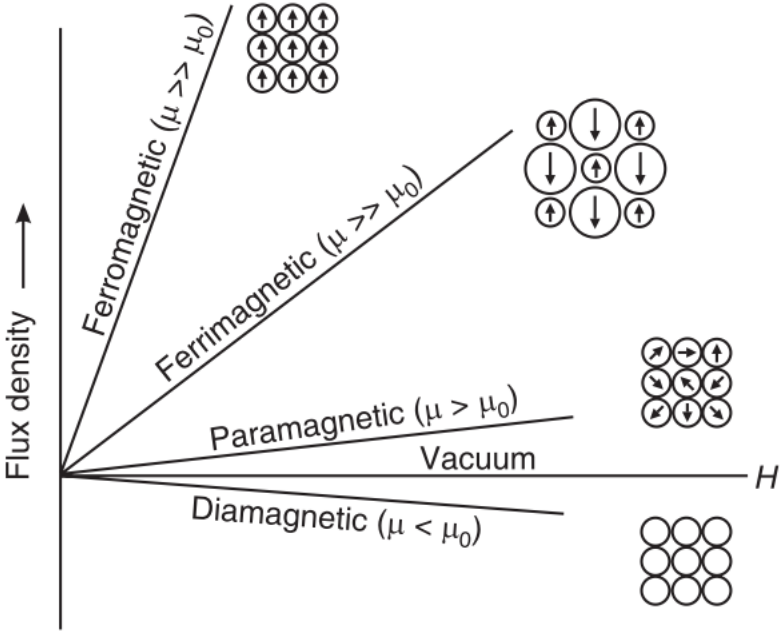
- The orbital motion of electrons
- The electron spin
- The nuclear spin

Paramagnetism

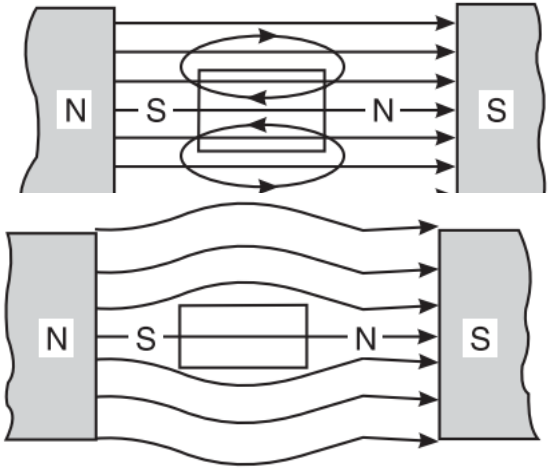
- Change in orbital motion in applied field

↓

Diamagnetism



DIAMAGNETIC MATERIALS

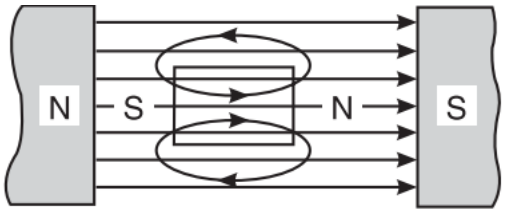


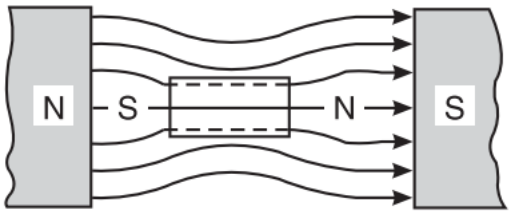
Hydrogen, air, water, gold, silver

1. Diamagnetic materials exhibit negative magnetic susceptibility (~ 10^{-6})
2. μ_r is slightly less than 1
3. If a diamagnetic body is placed in an inhomogeneous magnetic field, it tends to be pushed into the regions of weaker field.
4. magnetic susceptibility of diamagnetic materials is independent of temperature
5. M is a linear function of magnetic field H

PARAMAGNETIC MATERIALS

when placed in a magnetic field acquire feeble magnetism

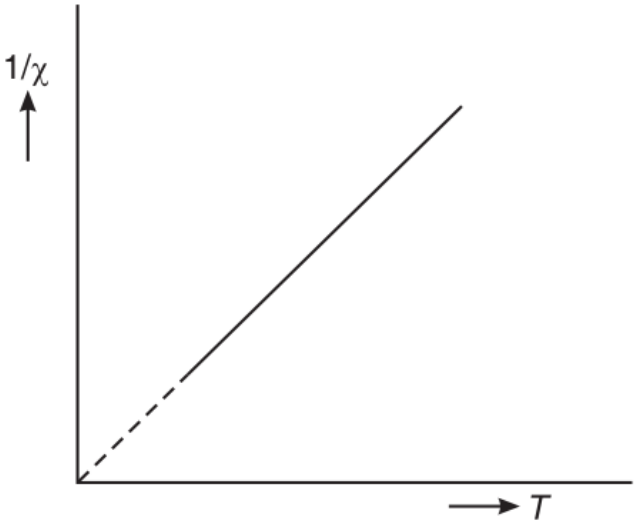




1. Paramagnetic materials exhibit positive magnetic susceptibility as the magnetization coincides in direction with the magnetic field H.

2. μ_r slightly higher than 1

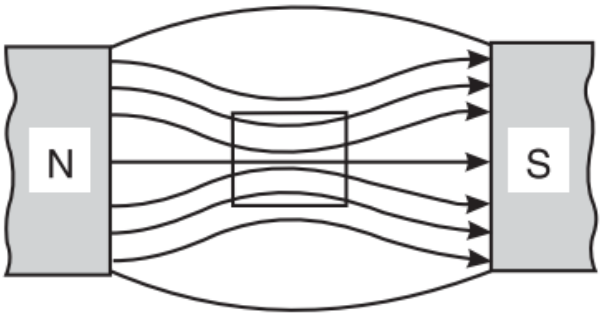
The paramagnetic susceptibility is strongly dependent on temperature

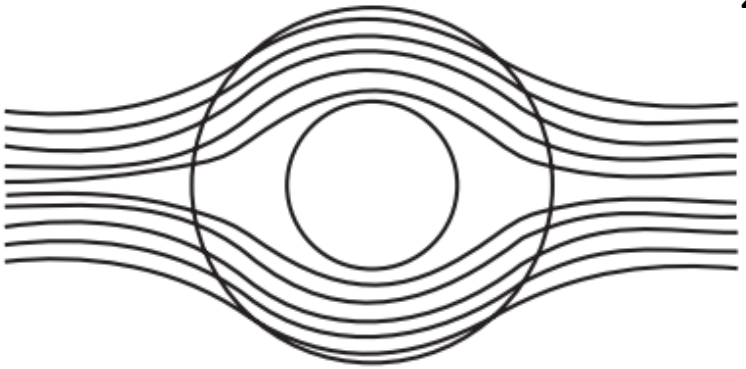


$$\chi_{para} = \frac{C}{T}$$ ➔ Curie Law

C = Curie Constant

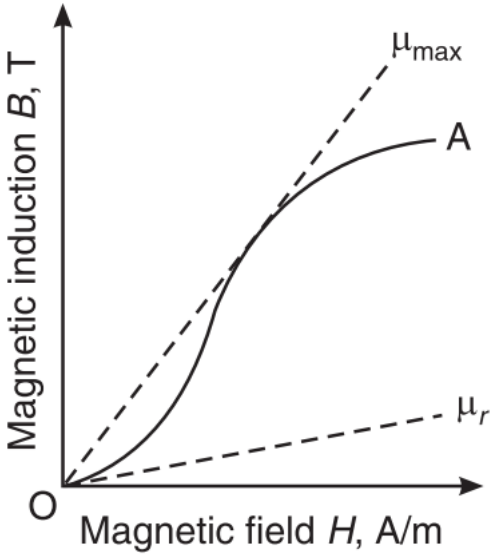
FERROMAGNETIC MATERIALS

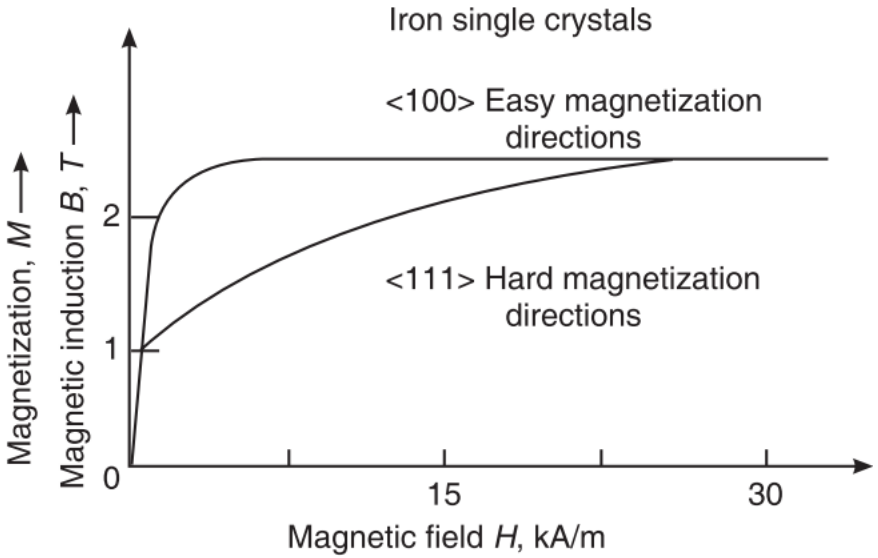




1. Ferromagnetic materials exhibit very high values of magnetic susceptibility and relative permeability

2. Ferromagnetic material attracts magnetic field lines





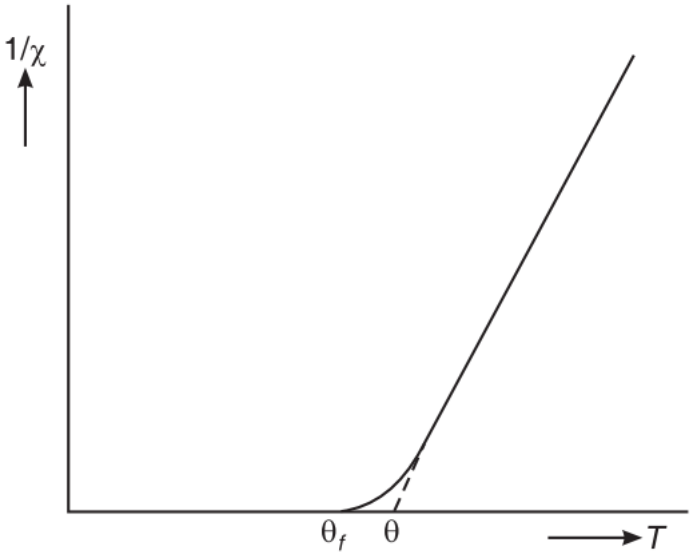
Total internal field in FM material

$H_i = H + \gamma M$ γ = Molecular field constant or Weiss constant

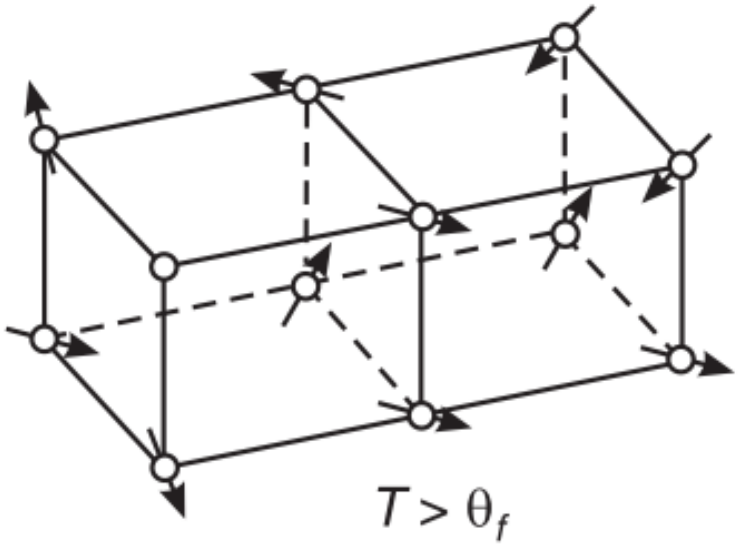
$$\chi_m = \frac{C}{T} = \frac{M}{H}$$

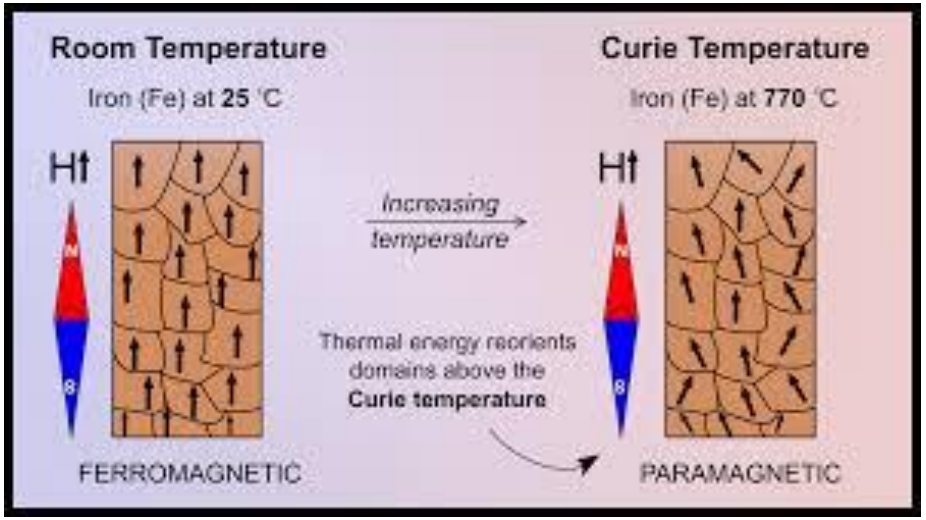
$$\therefore \chi_m = \frac{M}{H + \gamma M}$$

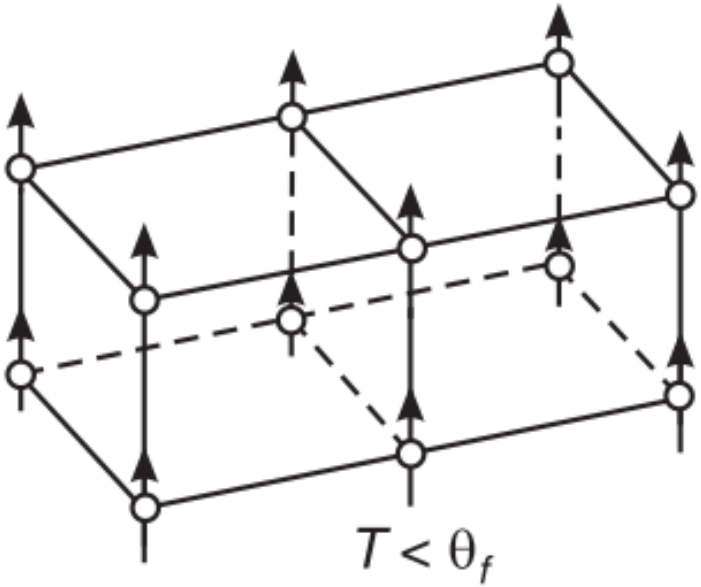
$$\rightarrow \frac{M}{H} = \frac{C}{T - \gamma C} = \frac{C}{T - \theta_f}$$



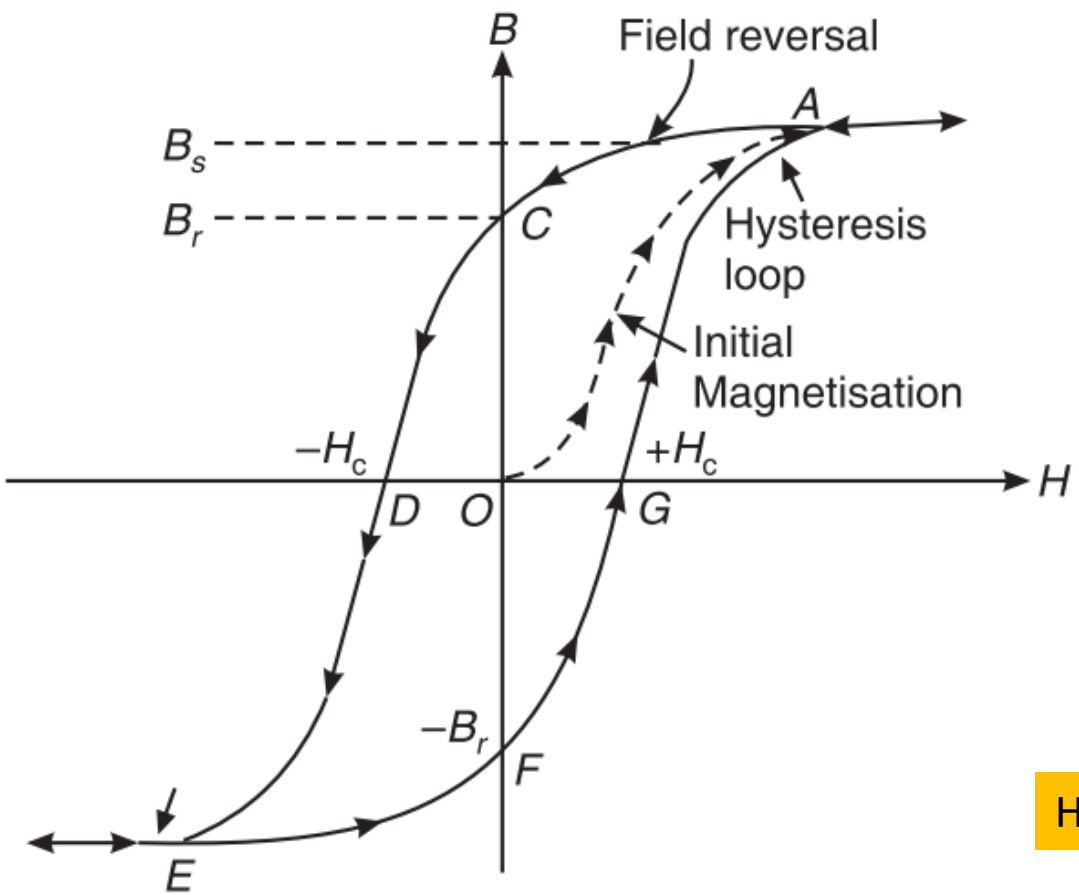
@physicsfun

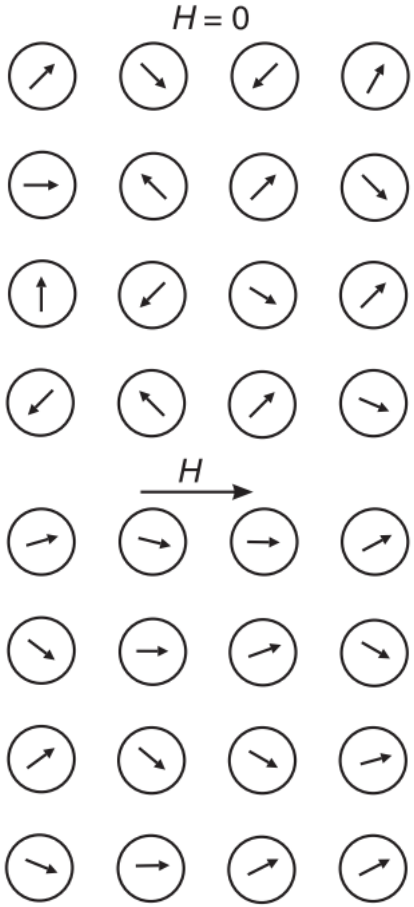

$T > \theta_f$


Room Temperature
Iron (Fe) at 25 °C
Curie Temperature
Iron (Fe) at 770 °C
H
Increasing temperature
Thermal energy reorients domains above the Curie temperature
FERROMAGNETIC
PARAMAGNETIC


$T < \theta_f$

Hysteresis





HYSTERESIS LOSS

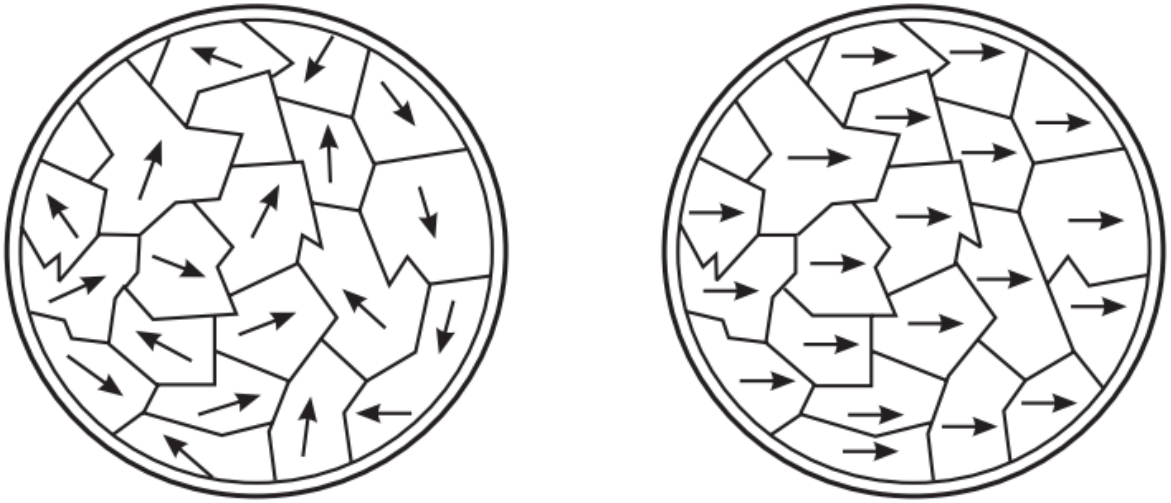
$\iint H\, dB\ (j/m^3)$ = Area of the Hysteresis loop

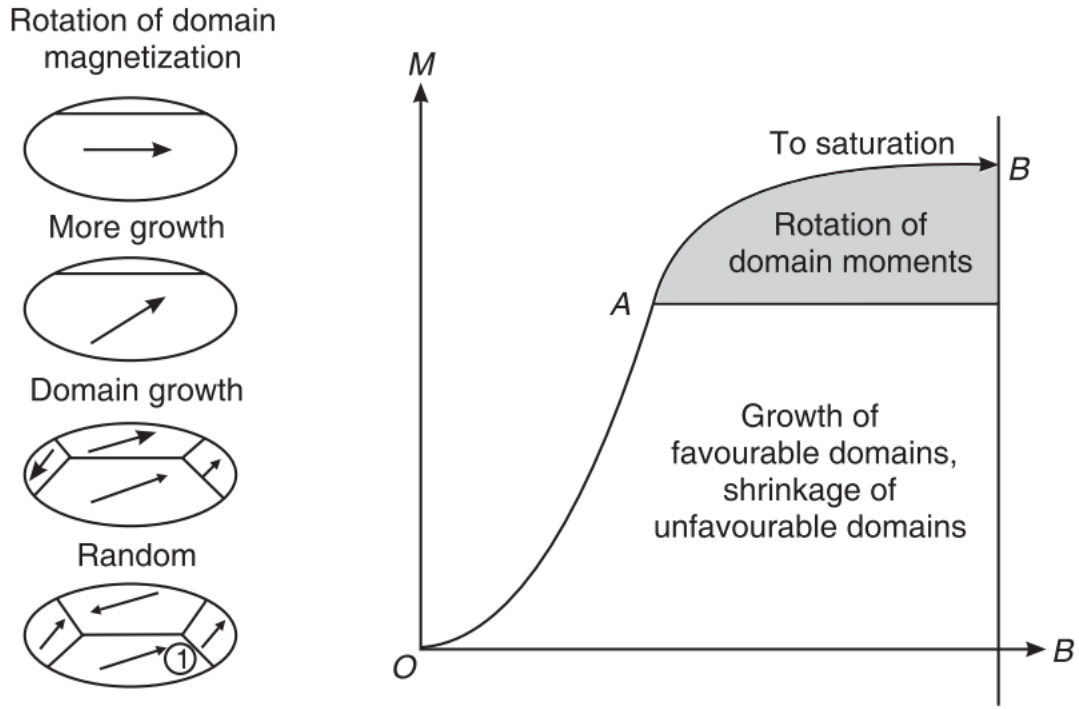
Bohr Magneton

$$\mu_B = \frac{eh}{4\pi m} = 9.28 \times 10^{-12}\ A.m^2$$

The susceptibility of paramagnetic $FeCl_3$ is 3.7×10^{-3} at 227°C. What will be the value of its susceptibility at -73°C and 27°C?

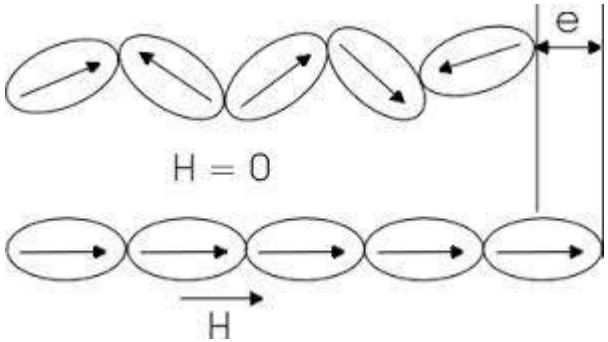
Magnetic Domains



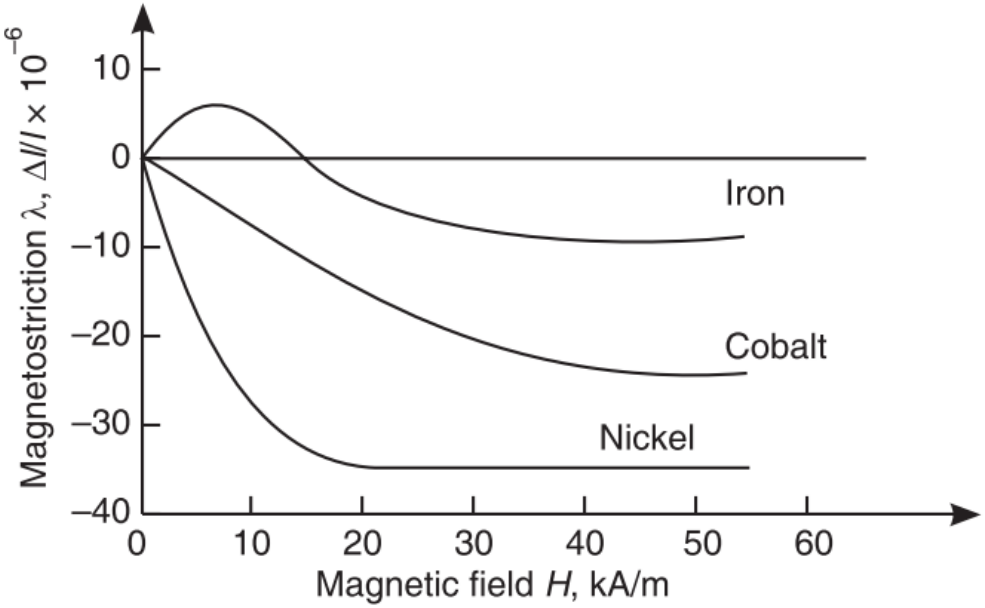


Process of magnetization: Using domain picture

MAGNETOSTRICTION



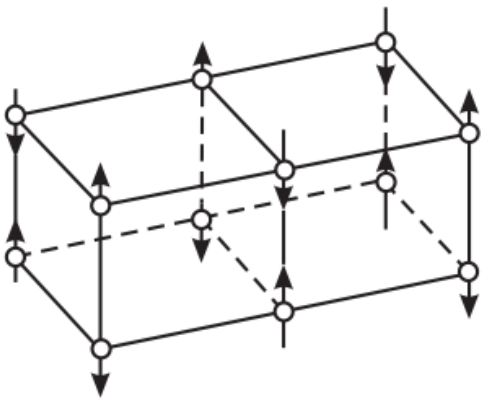
Change in volume of the material with magnetization

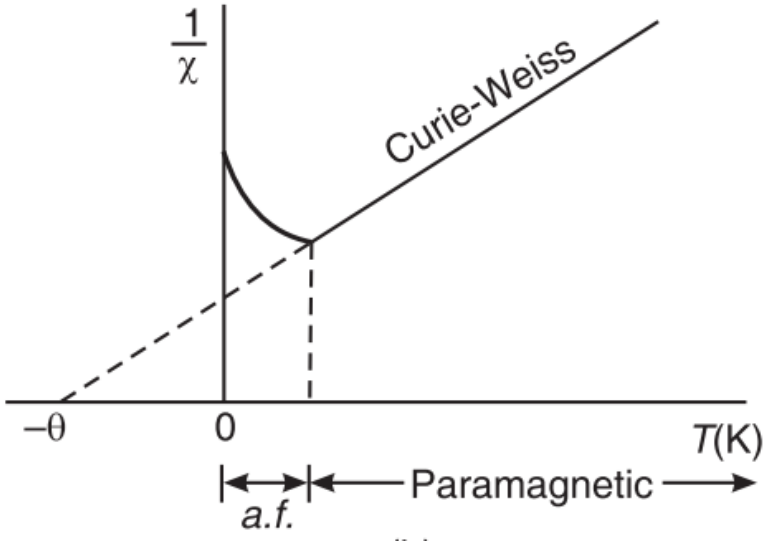


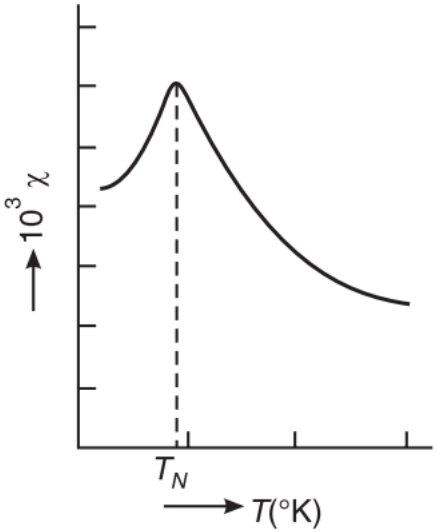
Ferromagnetics	Paramagnetics	Diamagnetics
Solids and possess crystalline structure	Solid, liquid or gas	Solid, liquid or gas
Strongly attracted towards magnetic field	Feeble attraction towards magnetic field	Feeble repulsion from the magnetic field
Field lines are concentrated in the material	More number of field lines pass through the material than outside	Less number of field lines pass through the material than outside
Set along the direction of the magnetic field	Tend to align along the magnetic field direction	Tend to align perpendicular to the magnetic field direction
Susceptibility, χ is large and positive	Susceptibility, χ is <1 but positive	Susceptibility, χ is small but negative
Relative permeability, μ_r is greater than unity	Relative permeability, μ_r is slightly greater than unity	Relative permeability, μ_r is less than unity
Susceptibility, χ decreases with temperature in a complex manner	Obeys Curie law, i.e. $\chi = \frac{1}{T}$	χ is independent of temperature

Have definite Curie point above which they become paramagnetic	No Curie point	No Curie point
B and M vary with H but not linearly and ultimately attain saturation	B and M vary with H linearly at low temperature and at high field tend towards saturation	B and M vary with H linearly but no saturation is reached
Exhibit phenomenon of Hysteresis	Hysteresis is not exhibited	Hysteresis is not exhibited
Possess retentivity	No retentivity	No retentivity
Iron, steel, cobalt, nickel	Platinum, Chromium, Aluminium, salts of Fe & Ni, oxygen	Bismuth, Mercury, Silver, Copper, water, air

ANTIFERROMAGNETISM



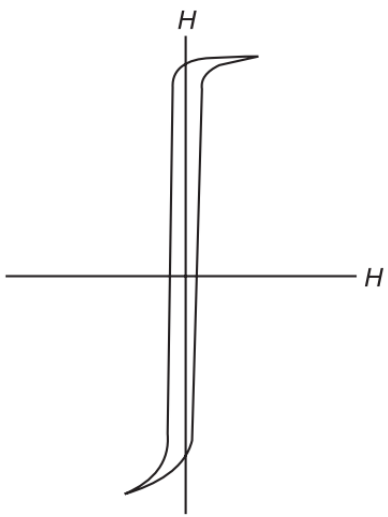




$$\chi = \frac{C}{T + \theta_C}$$

Mn and Cr are antiferromagnets

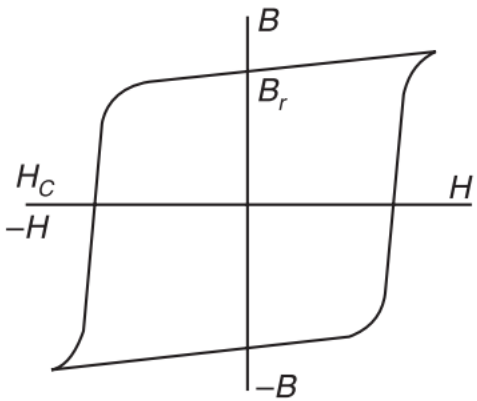
Soft Magnetic Materials



1. low remanence, low permeability, a low coercive field
2. Domain walls are easily movable
3. Free from impurity and defects

Use: power transformers, output transformers, motors, generators, electromagnets

Hard Magnetic Materials



1. high remanence, high permeability, a high coercive field and a large hysteresis loop

Use: permanent magnets

Soft Magnetic Materials

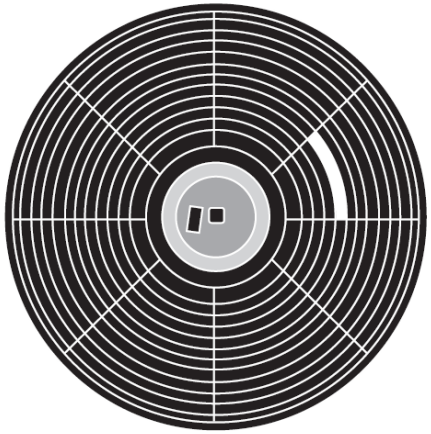
- Low Carbon Steel (0.05% C)
- Iron-silicon alloys (3 – 4 % Si)
- Nickel-iron alloys
- Mumetal (77% Ni,16% Fe, 5% Cu, 2% Cr)

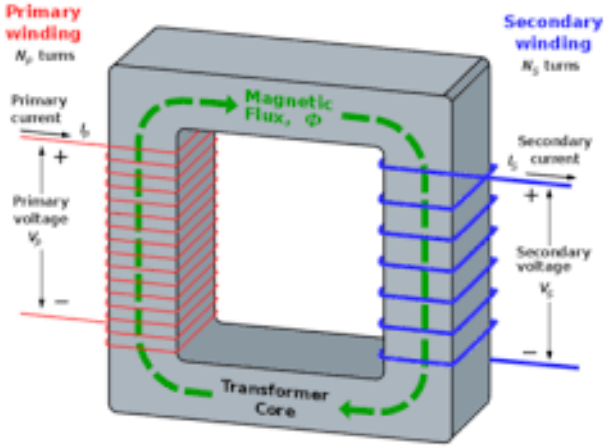
Hard Magnetic Materials

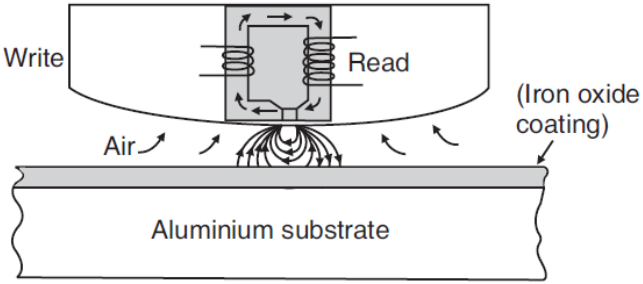
- Alnico alloys(Al, Ni, Co, Fe, Ti)
- Ferrites

Magnetic materials Applications

- Transformer Cores
- Memory Devices
- Sensors







Magnetic materials
Problems

1. Find the relative permeability of a ferromagnetic material, if a magnetic field strength 220 A/m produces a magnetization of 3300 A/m in it.

2. A magnetic material has a magnetization of 2300 A/m and produces a flux density of 0.00314 Wb/m^2. Calculate magnetizing force and relative permeability of the material.